

TAI NGUYEN PHU

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EDUCATION

University of Information Technology (UIT - VNUHCM)

Bachelor of Computer Science

Sep 2022 - Sep 2025

GPA: 3.3/4.0

Le Khiet High School for the Gifted

Information Technology

Sep 2019 - Jun 2022

GPA: 3.4/4.0

ACHIEVEMENTS

- **IELTS Academic:** 7.0 Overall [🔗](#)
- **UIT Global Scholarship** [🔗](#)
- **Kaggle Competition:** "Credit Risk Model Stability": Rank: 1096/3858 [🔗](#)

SKILLS

Languages: Proficient in Python, C & C++; Basic in Javascript, SQL, HTML, CSS

Deployment Tools: Git, GitHub Actions, Docker, AWS, FastAPI, Weight & Bias

ML/DL Frameworks: PyTorch, Transformers, PEFT, Chroma, FAISS, OpenCV, Numpy, Pandas, Scikit-learn, Spark

Core AI: HuggingFace, LangChain, LangSmith, LangGraph, smolagents, Chroma

Mathematics: Linear algebra, Discrete math, Calculus, Statistics and Probability, Data structure and algorithm.

PROJECTS

RAG Chatbot with Memory and Multimodal-Multilingual Support [🔗](#)

May 2025 - July 2025

- Built a RAG system for scientific PDF files that enables multimodal processing capabilities on text, table, and image data.
- Designed a multilingual, multimodal retrieval pipeline using BGE-M3 embeddings and Chroma vector databases.
- Conducted prompt engineering with multi-turn prompts and scenario-based testing using the Gemini Model API.
- Implemented conversational memory management using LangChain's ConversationSummaryMemory, enabling context-aware follow-up QA in chatbot interactions.
- Implemented a citation-aware abstractive QA mechanism, mapping answers to source documents with text-level references for verifiability.
- Developed an arXiv paper search engine, allowing dynamic retrieval of 150,000+ scientific publications in real-time.
- Containerized and deployed a FastAPI-based end-to-end web application on AWS EC2 and Hugging Face Spaces for reproducibility and scalability.

Vietnamese Legal Document Semantic Retrieval System [🔗](#)

Mar 2025 - Apr 2025

- Fine-tuned a multilingual SBERT model on a curated Vietnamese legal corpus to improve domain-specific semantic retrieval accuracy.
- Achieved NDCG@10: 60.4% and MAP@10: 53.6% on the MTEB benchmark (BKAI Legal Retrieval dataset).
- Built a full semantic retrieval pipeline: data preprocessing, model fine-tuning (contrastive learning), evaluation using MTEB, and deployment.
- Implemented a GPU-accelerated FAISS index with approximately 100K documents for ANN search with sub-second latency.
- Deployed an end-user interface using Gradio, containerized via Docker for scalable and reproducible deployment.

Glassdoor 2024 Data Science Job Salary Prediction [🔗](#)

Dec 2024 - Jan 2025

- Developed a machine learning-powered salary prediction system trained on Glassdoor 2024 Data Science job listings.
- Conducted comprehensive EDA and feature engineering, including data preprocessing, categorical encoding, missing value imputation, and creation of interaction terms to capture complex salary dynamics.
- Achieved $R^2 = 0.815$ with XGBoost, significantly outperforming the baseline Linear Regression ($R^2 = 0.7$).
- Performed hyperparameter optimization via GridSearchCV to maximize model performance.
- Built and deployed an Streamlit web application with real-time salary predictions based on user-provided job attributes.